GRAPH THEORY LADDER 3 OF 9

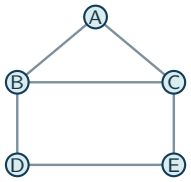
Adjacency matrices

A graph becomes a table of zeros and ones, and the powers of that table count walks.

1 / THE RULE AND THE PICTURE

The entry in row u and column v is 1 when u and v are joined

	A	B	C	D	E
A	0	1	1	0	0
B	1	0	1	1	0
C	1	1	0	0	1
D	0	1	0	0	1
E	0	0	1	1	0



Listing the vertices in the order A, B, C, D, E, the matrix M has $M_{uv} = 1$ for an edge and 0 otherwise, and the key property is

$$(M^k)_{uv} = \text{number of walks of length } k \text{ from } u \text{ to } v$$

because a walk of length $k+1$ is a walk of length k followed by one more edge.

2 / READING THE MATRIX

Symmetry, row sums and the diagonal

The matrix is symmetric, since an edge joins u to v exactly when it joins v to u, and the diagonal is zero, since there are no loops. Each row sum is a degree, so the row sums of the house matrix are $2, 3, 3, 2, 2$, and the sum of all entries is $12 = 2E$, the handshake rule. For a directed graph, such as web links, the matrix is no longer symmetric and the rows count outgoing links.

3 / POWERS COUNT WALKS

The squares and cubes of the house matrix

$$M^2 = \begin{pmatrix} 2 & 1 & 1 & 1 & 1 \\ 1 & 3 & 1 & 0 & 2 \\ 1 & 1 & 3 & 2 & 0 \\ 1 & 0 & 2 & 2 & 0 \\ 1 & 2 & 0 & 0 & 2 \end{pmatrix}, \qquad M^3 = \begin{pmatrix} 2 & 4 & 4 & 2 & 2 \\ 4 & 2 & 6 & 5 & 1 \\ 4 & 6 & 2 & 1 & 5 \\ 2 & 5 & 1 & 0 & 4 \\ 2 & 1 & 5 & 4 & 0 \end{pmatrix}.$$

In M^2 the diagonal gives the degrees $2, 3, 3, 2, 2$, because a walk of length 2 from u back to u goes out along an edge and returns along it, and the entry $(B, E) = 2$ counts the walks B, C, E and B, D, E. In M^3 the entry $(A, B) = 4$ counts four walks of length 3 from A to B, and the sum of all entries of M^2 is $30 = \sum (\deg v)^2$.

4 / TRIANGLES AND EIGENVALUES

Closed walks of length 3 are triangles

The trace of M^3 is 6, which is 6 times the number of triangles, because each triangle gives closed walks of length 3 from three starting points in two directions, and the house graph has one triangle.
The eigenvalues of M are $2.481,\ 0.689,\ 0,\ -1.170,\ -2$, and the largest, 2.481, exceeds the average degree $12/5 = 2.4$, as it always does, while the eigenvalues are what let us study large graphs without drawing them.

5 / TRY THESE

Use the table, not the picture

(a) Which vertex has the largest row sum in the house matrix? B and C with degree 3.
(b) How many walks of length 2 join A to D? Row A, column D of M^2 is 1, the walk A, B, D.
(c) How many walks of length 2 go from A to A? 2, which is the degree of A.

ASIDE / ADJACENCY LISTS

For a large sparse graph a list of neighbours uses far less space than a matrix with mostly zeros, though the matrix makes the algebra possible.

LINKS TO OTHER SHEETS

The linear algebra ladder explains matrix powers and eigenvalues, and the Laplacian sheet later adjusts this matrix by the degrees.

Convention. M adjacency matrix with vertices ordered A, B, C, D, E; $(M^k)_{uv}$ counts walks of length k; the trace is the sum of the diagonal.